Complex Eigenvalues and Eigenvectors Reading

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Up until now we have only considered eigenvalues and eigenvectors made up of real numbers. However, there are characteristic polynomials that have roots that are not real numbers. For instance, the matrix

$$A = \begin{bmatrix} 3 & -4 \\ 1 & 3 \end{bmatrix}$$

has characteristic polynomial

$$|A - \lambda I_2| = (3-\lambda)^2 + 4$$

This characteristic polynomial has no real roots, but it has two complex roots. Eigenvalues and eigenvectors involving complex numbers have useful applications, which we will get to later in this lesson and in the next. We start here with a brief review of the properties of complex numbers. Feel free to skip to the next subsection if you are already familiar with the basics of complex numbers.

Complex Numbers

We are not going to fully develop the complex numbers, but instead just focus on the aspects that are needed later. If a and b are real numbers, then a typical complex number has the form

$z = a + ib$

where i satisfies $i^2 = -1$, making i a square root of -1. Here a is called the real part of z, denoted by $\text{Re}(z)$, and b is the imaginary part, denoted by $\text{Im}(z)$. (Note that both the real and imaginary parts are real numbers.) The set of all complex numbers is denoted by $\mathbf{C}$.

To add complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, we just add the real parts and the imaginary parts separately,

$$z_1 + z_2 = (a_1 + ib_1) + (a_2 + ib_2) = (a_1 + a_2) + i(b_1 + b_2)$$

For example, if $z_1 = 3 + i$ and $z_2 = 2 + 3i$, then

$$z_1 + z_2 = (3 + i) + (2 + 3i) = (3 + 2) + i(1 + 3) = 5 + 4i$$

Adding complex numbers is similar to adding vectors in $\mathbf{R}^2$, with each component added separately.

We can also represent complex numbers geometrically just as we do vectors in $\mathbf{R}^2$, with $\text{Re}(z)$ on the x-axis and $\text{Im}(z)$ on the y-axis. For example, z_1 and z_2 from above are shown in Figure 1.



Figure 1 $z_1 = 3 + i$ and $z_2 = 2 + 3i$.

The product of complex numbers is found by multiplying term by term and then simplifying using the identity $i^2 = -1$.

$$\begin{aligned} z_1 z_2 &= (a_1 + ib_1)(a_2 + ib_2) \\ &= a_1a_2 + ia_1b_1 + ia_2b_1 + i^2b_1b_2 = (a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1) \end{aligned}$$

For our two complex numbers $z_1 = 3 + i$ and $z_2 = 2 + 3i$, we have

$$z_1 z_2 = (3 + i)(2 + 3i) = ((3)(2) - (1)(3)) + i((3)(3) + (2)(1)) = 3 + 11i$$

An alternate way to represent a complex number is to use polar coordinates. Define the modulus of a complex number $z = a + ib$ by

$$|z| = \sqrt{a^2 + b^2}.$$

The modulus extends the absolute value to complex numbers and gives the distance from z to the origin. The argument of z, denoted by $\arg(z)$, is the angle θ (in radians) in the counter clockwise direction from the positive x-axis to the ray from the origin to z (see Figure 2). Note that the argument is not unique, because we can always add or subtract multiples of 2π.

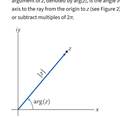


Figure 2 $|z|$ and $\arg(z)$.

If $r = |z|$ and $\theta = \arg(z)$, then we can express z in polar form as

$$z = r(\cos(\theta) + i\sin(\theta))$$

For instance, if $r = 5$ and $\theta = \pi/3$, then we can convert to rectangular coordinates by evaluating, with

$$z = 5(\cos(\pi/3) + i\sin(\pi/3)) = \tfrac{5}{2} + i\tfrac{5\sqrt{3}}{2}$$

Converting from rectangular to polar coordinates is depicted in Figure 3. For $z = a + ib$, we set $r = |z| = \sqrt{a^2 + b^2}$ and have

$$\tan(\theta) = \tfrac{b}{a}$$

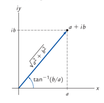


Figure 3 Converting from rectangular to polar coordinates.

For example, in the case of $z_1 = 3 + i$, we have

$$r = \sqrt{3^2 + 1^2} = \sqrt{10} \quad \text{and} \quad \theta = \tan^{-1}\left(\tfrac{1}{3}\right) \approx 0.3218 \text{ radians}$$

An interesting formula arises when multiplying complex numbers written in polar form. If

$$z_1 = r_1(\cos(\theta_1) + i\sin(\theta_1)) \quad \text{and} \quad z_2 = r_2(\cos(\theta_2) + i\sin(\theta_2))$$

then we have

$$\begin{aligned} z_1 z_2 &= r_1(\cos(\theta_1) + i\sin(\theta_1)) \cdot r_2(\cos(\theta_2) + i\sin(\theta_2)) \\ &= r_1 r_2\{(\cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2)) + i(\cos(\theta_1)\sin(\theta_2) + \cos(\theta_2)\sin(\theta_1))\} \\ &= r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)) \end{aligned}$$

with the last line following from trigonometric identities. This formula tells us that

$$|z_1 z_2| = |z_1||z_2| \quad \text{and} \quad \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

Applying this repeatedly (by induction) to $z = r(\cos\theta) + i\sin\theta)$, we find that for each positive integer k,

$$z^k = r^k(\cos(k\theta) + i\sin(k\theta))$$

In the special case where $z = \cos\theta) + i\sin\theta)$ (that is, $r = 1$), this yields *DeMoivre's Formula*,

$$(\cos(\theta) + i\sin(\theta))^k = \cos(k\theta) + i\sin(k\theta)$$

For $z = a + ib$, the exponential function e^z extends to the complex numbers by the definition

$$e^{a+ib} = e^a(\cos(b) + i\sin(b))$$

Note that if $z = a$ is real, then $e^z = e^a$ reduces to the usual exponential function. On the other hand, if $z = ib$ is purely imaginary, then

$$e^{ib} = \cos(b) + i\sin(b)$$

and $|e^{ib}| = 1$.

The complex conjugate of $z = a + ib$ is given by $\overline{z} = a - ib$. Complex conjugation distributes across addition and multiplication, so that if z and w are complex numbers, then

$$\overline{z + w} = \overline{z} + \overline{w} \quad \text{and} \quad \overline{zw} = \overline{z} \cdot \overline{w}$$

One interesting consequence of these properties is that for polynomials with real coefficients, complex roots come in conjugate pairs. that is, if

$$f(z) = a_n z^n + \cdots + a_1 z + a_0$$

has real coefficients and $f(z_0) = 0$, then $f(\overline{z_0}) = 0$. To see why, note that $\overline{x} = x$ for any real number x. If $f(z_0) = 0$, then

$$\begin{aligned} 0 = \overline{0} = \overline{f(z_0)} &= \overline{a_n z_0^n + \cdots + a_1 z_0 + a_0} \\ &= \overline{a}_n \overline{z_0^n} + \cdots + a_1 z_0 + a_0 \\ &= a_n(\overline{z_0})^n + \cdots + a_1 \overline{z_0} + a_0 = f(\overline{z_0}) \end{aligned}$$

which tells us that $f(\overline{z_0}) = 0$ as well. For example, if $f(z) = z^2 + 2z + 4$, then the solutions to $f(z) = 0$ can be found by applying the quadratic formula,

$$z = \tfrac{-2 \pm \sqrt{2^2 - 4(1)(4)}}{2(1)} = \tfrac{-2 \pm \sqrt{-12}}{2} = -1 \pm i\sqrt{3}$$

Therefore we have two solutions, $z = -1 + i\sqrt{3}$ and the conjugate $\overline{z} = -1 - i\sqrt{3}$.

One benefit of expanding from the real numbers to the complex numbers is that we can completely factor polynomials. Specifically, any polynomial $f(z)$ of degree n with real or complex coefficients can be factored completely to

$$f(z) = c(z - z_1)(z - z_2)\cdots(z - z_n)$$

where $c, z_1, \ldots z_n$ are complex numbers.

Given an $n \times n$ matrix A, we know that the characteristic polynomial will have degree n. When working in the complex numbers, the characteristic polynomial has exactly n roots (counting multiplicities), so that A must have exactly n eigenvalues (again, counting multiplicities).

Complex Eigenvalues and Eigenvectors

Now that we have refreshed our knowledge of complex numbers, let's return to our opening problem.

Example 1

Find the eigenvalues and associated eigenvectors for the matrix

$$A = \begin{bmatrix} 3 & -4 \\ 1 & 3 \end{bmatrix}$$

Solution

We know that $\det(A - \lambda I_2) = (3 - \lambda)^2 + 4$. Setting this equal to 0 and solving for λ, we have

$$(3 - \lambda)^2 = -4 \quad \Rightarrow \quad 3 - \lambda = \pm\sqrt{-4} \pm 2i \quad \Rightarrow \quad \lambda = 3 \pm 2i$$

In general, it is algebraically messy to find complex eigenvectors by hand. However, it is manageable for 2×2 matrices. For the eigenvector $\lambda_1 = 3 - 2i$, we have

$$A - \lambda_1 I_2 = A - (3 - 2i) I_2 = \begin{bmatrix} 2i & -4 \\ 1 & 2i \end{bmatrix}$$

The augmented matrix of $(A - \lambda_1 I_2)\mathbf{u} = \mathbf{0}$ and the corresponding echelon form are

$$\left[\begin{array}{cc|c} 2i & -4 & 0 \\ 1 & 2i & 0 \end{array}\right] \underset{\sim}{\left[-\tfrac{1}{2}iR_1 + R_2 \to R_2\right]} \left[\begin{array}{cc|c} 2i & -4 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

The echelon form is equivalent to the equation $2iu_1 - 4u_2 = 0$. A nontrivial solution is $u_1 = 2$ and $u_2 = i$, which gives us the eigenvector $\mathbf{u}_1 = \begin{bmatrix} 2 \\ i \end{bmatrix}$. Similar calculations applied to $\lambda_2 = 3 + 2i$ can be used to produce the associated eigenvector $\mathbf{u}_2 = \begin{bmatrix} 2 \\ -i \end{bmatrix}$.

For a vector $\mathbf{z}$, the complex conjugate $\overline{\mathbf{z}}$ means that we take the complex conjugate for each entry of $\mathbf{z}$. An analogous definition holds for $\overline{A}$, the complex conjugate of the matrix A.

The eigenvalues in Example 1 are a conjugate pair, with $\lambda_2 = \overline{\lambda_1}$. The corresponding eigenvectors are similarly related, with $\mathbf{u}_2 = \overline{\mathbf{u}_1}$. This is true for any square matrix with real entries.

THEOREM 13

Suppose that A is a real matrix with eigenvalue λ and associated eigenvector $\mathbf{u}$. Then $\overline{\lambda}$ is also an eigenvalue of A, with associated eigenvector $\overline{\mathbf{u}}$.

Proof

If A is a real matrix, then the characteristic polynomial has real coefficients. Previously we showed that complex roots of polynomials with real coefficients come in conjugate pairs. Thus, if λ is a complex eigenvalue of a matrix A, then so is $\overline{\lambda}$.

Next suppose that $\mathbf{u}$ is an eigenvector of A associated with λ. Since A has real entries, we have $A = \overline{A}$, so that

$$A\overline{\mathbf{u}} = \overline{A}\overline{\mathbf{u}} = \overline{A\mathbf{u}} = \overline{\lambda\mathbf{u}} = \overline{\lambda}\overline{\mathbf{u}}$$

Hence $\overline{\mathbf{u}}$ is an eigenvector of A associated with eigenvalue $\overline{\lambda}$.

Example 2

Find the eigenvalues and associated eigenvectors for the matrix

$$A = \begin{bmatrix} -1 & 3 & -4 \\ -2 & 3 & -4 \\ 1 & 1 & 3 \end{bmatrix}$$

Solution

Starting with the characteristic polynomial, we have

$$\det(A - \lambda I_3) = -\lambda^3 + 5\lambda^2 - 17\lambda + 13 = -(\lambda - 1)(\lambda^2 - 4\lambda + 13)$$

Thus one eigenvalue is $\lambda_1 = 1$. Applying the quadratic formula to the quadratic term shows the other eigenvalues are $\lambda_2 = 2 + 3i$ and $\lambda_3 = \overline{\lambda_2} = 2 - 3i$.

For the eigenvectors associated with $\lambda_1 = 1$, the augmented matrix and corresponding echelon form are

$$\left[\begin{array}{ccc|c} -2 & 3 & -4 & 0 \\ -2 & 2 & -4 & 0 \\ 1 & 1 & 2 & 0 \end{array}\right] \underset{\sim}{\begin{array}{l}[-R_1 + R_2 \to R_2 \\ [R_1 + R_3 \to R_3 \\ [R_2 + R_3 \to R_3\end{array}} \left[\begin{array}{ccc|c} -2 & 3 & -4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The row operations are

$-\frac{1}{2}R_1 + R_2 \to R_2$,
$\frac{1}{2}R_1 + R_3 \to R_3$,
$-\frac{1}{4}R_2 + R_3 \to R_3$

Back substitution yields the eigenvector $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$. For $\lambda_2 = 2 + 3i$, we have

$$\left[\begin{array}{ccc|c} (-3-3i) & 3 & -4 & 0 \\ -2 & (1-3i) & -4 & 0 \\ 1 & 1 & (1-3i) & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} (-3-3i) & 3 & -4 & 0 \\ 0 & -2i & -(8+4i)/3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

After back substitution and scaling we find that $\mathbf{u}_2 = \begin{bmatrix} -1 + 5i \\ -2 - 4i \\ 3 \end{bmatrix}$.

There is no need for row operations to find $\mathbf{u}_3$. Since $\lambda_3 = \overline{\lambda_2}$, we know from Theorem 13 that

$\mathbf{u}_3 = \overline{\mathbf{u}_2} = \begin{bmatrix} -1 - 5i \\ -2 + 4i \\ 3 \end{bmatrix}$.

Rotation-Dilation Matrices

One application of complex eigenvalues and eigenvectors is in analyzing the behavior of a special class of 2×2 matrices. For illustrative purposes we distribute eight vectors $\mathbf{x}_1, \ldots, \mathbf{x}_8$ evenly around the unit circle as shown in Figure 4(a). Now define the matrix

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

Figure 4(b) shows the vectors $A\mathbf{x}_1, \ldots, A\mathbf{x}_8$.

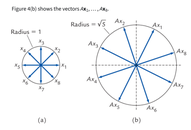


Figure 4 (a) The vectors $\mathbf{x}_1, \ldots, \mathbf{x}_8$ are evenly spaced around the unit circle. (b) The vectors $\mathbf{x}_1, \ldots, \mathbf{x}_8$ after multiplication by A, which rotates each by 63.43° and dilates each by $\sqrt{5}$. (NOTE: Figures are not drawn to scale.)

Comparing $\mathbf{x}_1$ with $A\mathbf{x}_1$, we see that multiplication by A causes each $\mathbf{x}_i$ to be rotated and dilated by the same amount. It turns out that the same thing will happen for any vector $\mathbf{x}$ when multiplied by A.

This rotation -dilation behavior happens with any real matrix of the form

$$A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

To see why, first note that if $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, then

$$A\mathbf{x} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} ax_1 - bx_2 \\ bx_1 + ax_2 \end{bmatrix}$$

It can be shown that $\lambda = a + ib$ is an eigenvalue of A. Next note that if we let $x = x_1 + ix_2$, then

$$\lambda x = (a + ib)(x_1 + ix_2) = (ax_1 - bx_2) + i(bx_1 + ax_2)$$

Thus the components of $A\mathbf{x}$ match the real and imaginary parts of λx. If we think of the real and imaginary parts as components of λx, then the products $A\mathbf{x}$ and λx can be viewed as equivalent. From the properties of the products of complex numbers, we know that

$$\begin{array}{ll} \arg(\lambda x) = \arg(\lambda) + \arg(x) & \text{(Rotation of } x \text{ in } C \text{ by the angle } \arg(\lambda)) \\ |\lambda x| = |\lambda||x| = \sqrt{a^2 + b^2}\,|x| & \text{(Dilation of } x \text{ in } C \text{ by the multiple } |\lambda|) \end{array}$$

Therefore the eigenvalue tells us the amount of rotation and dilation induced by A. Returning to our matrix

$$A = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

we have $\lambda = 1 + 2i$, so that $A\mathbf{x}$ will produce

$$\begin{array}{l} \text{Rotation by } \arg(\lambda) = \tan^{-1}(2/1) \approx 63.43 \text{ degrees} \\ \text{Dilation by } |\lambda| = \sqrt{1^2 + 2^2} = \sqrt{5} \end{array}$$

Note that this is consistent with Figure 4.

Example 3

Determine the rotation and dilation that result from multiplying $\mathbf{x}$ in $\mathbf{R}^2$ by

$$A = \begin{bmatrix} 7 & -4 \\ 4 & 7 \end{bmatrix}$$

Solution

An eigenvalue of A is $\lambda = 7 + 4i$, so that we have

$$\begin{array}{l} \text{Rotation by } \tan^{-1}(4/7) \approx 0.5191 \text{ radians} \\ \text{Dilation by } \sqrt{7^2 + 4^2} = \sqrt{65} \end{array}$$

The Hidden Rotation–Dilation Matrix

As it happens, *any* 2×2 real matrix with complex eigenvalues has a rotation–dilation hidden within it. Finding this rotation–dilation requires a procedure reminiscent of diagonalization. We start by illustrating using the matrix from the beginning of this lesson,

$$A = \begin{bmatrix} 3 & -4 \\ 1 & 3 \end{bmatrix}$$

We have previously shown that $\mathbf{u} = \begin{bmatrix} 2 \\ i \end{bmatrix}$ is an eigenvector of A. Now form the matrix

$$P = \begin{bmatrix} \text{Re}(\mathbf{u}) & \text{Im}(\mathbf{u}) \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

where Re(**u**) and Im(**u**) denote the vectors formed by taking the real and imaginary parts of each component of **u**. Then we compute

$$P^{-1}AP = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix} = B$$

Thus $A = PBP^{-1}$, where B is a rotation–dilation matrix. Note that the first row of B corresponds to the real and imaginary parts of the eigenvalue $\lambda = 3 - 2i$ of A associated with $\mathbf{u}$. This example is generalized in the next theorem.

THEOREM 14

Let A be a nonzero real 2×2 matrix with complex eigenvalue $\lambda = a - ib$ and associated eigenvector $\mathbf{u}$. If $P = \begin{bmatrix} \text{Re}(\mathbf{u}) & \text{Im}(\mathbf{u}) \end{bmatrix}$, then

$$A = PBP^{-1}$$

where $B = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ is a rotation–dilation matrix.

We do not give a proof here. The form PBP^{-1} suggests viewing the transformation $A\mathbf{x}$ as the composition of transformations. The first is a change to the basis {Re(**u**), Im(**u**)}, the second is a rotation–dilation, and the third is a change back to the standard basis. Note that the rotation–dilation is not applied to $\mathbf{x}$, but to the coordinate vector of $\mathbf{x}$ relative to the basis {Re(**u**), Im(**u**)}.

Example 4

Find the hidden rotation–dilation matrix within

$$A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$$

Solution

The characteristic polynomial is $\det(A - \lambda I_2) = \lambda^2 - 4\lambda + 13$. A quick application of the quadratic formula reveals that one of the eigenvalues is $\lambda = 2 - 3i$. (What is the other?) The usual procedure yields an associated eigenvector $\mathbf{u} = \begin{bmatrix} 1 + 3i \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + i\begin{bmatrix} 3 \\ 0 \end{bmatrix}$. Applying Theorem 14 with $\lambda = 2 - 3i$, we have

$$P = \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix}$$

Clearly B has the form of a rotation–dilation matrix. We can check our calculations with the computation

$$PBP^{-1} = \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 3 & 2 \end{bmatrix} \left(-\tfrac{1}{6}\begin{bmatrix} 0 & -3 \\ -2 & 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix} = A$$

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